

AI Based Attendance Monitoring System

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Abstract

The AI Based Attendance Monitoring System is a Python-based desktop application that automates attendance using facial recognition. The system captures student face images, trains an LBPH face recognition model, and recognizes registered students through a webcam. Once a student is recognized, attendance is automatically recorded with the date and time in a CSV file.

Introduction

Attendance management is an important task in schools and colleges. Traditional attendance systems are time-consuming and can lead to errors or proxy attendance. This project uses face recognition technology to automatically identify students and record their attendance.

Objectives

- Automate attendance.
- Reduce manual work.
- Prevent proxy attendance.
- Improve accuracy.
- Store records digitally.

Software Requirements

Python 3.x, OpenCV, NumPy, Pandas, Pillow (PIL), Tkinter.

Hardware Requirements

Computer/Laptop, Webcam, 4 GB RAM, Windows OS.

Technologies Used

Python, OpenCV, Tkinter, NumPy, Pandas, Pillow.

Working

1. Enter student ID and name.
2. Capture face images.
3. Train the face recognition model.
4. Enter the password for registration.
5. Start attendance.
6. Attendance is saved automatically in a CSV file.

Advantages

Easy to use, saves time, automatic attendance, reduces human error, prevents proxy attendance, digital record keeping.

Limitations

Requires good lighting, depends on webcam quality, only registered students can be recognized.

Future Scope

Database integration, cloud attendance, mobile application, deep learning recognition, web portal.

Conclusion

The system automates attendance using facial recognition, reduces manual effort, improves accuracy, and stores attendance records digitally.

References

Python Documentation, OpenCV Documentation, NumPy Documentation, Pandas Documentation, Pillow Documentation.